



WTiUE

**RANDOM-FOREST MODELLING OF LOCAL HEAT
TRANSFER COEFFICIENTS DURING FLOW BOILING
IN MINICHANNELS: A REPRODUCIBLE PYTHON
PIPELINE AND TIME-BLOCKED VALIDATION**

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This study presents a reproducible, Python-based pipeline for estimating and modelling local heat transfer coefficients (HTC) in minichannel flow boiling experiments. The workflow was applied to two datasets covering different working fluids and module orientations (horizontal multi-channel and vertical few-channel configurations). Local HTC was computed from Newton's law using a regime-dependent reference temperature. To handle uncertainty near saturation, a temperature-difference deadband was introduced to assign three labels (subcooled/ambiguous/saturated), with ambiguous samples excluded from training and reporting. Random-Forest regression was trained in log-space and evaluated using time-blocked cross-validation to avoid leakage across transient operating ramps. Under this scheme, saturated-regime predictions remained consistently accurate across stable time blocks (R^2 on $\log_{10}(\text{HTC}) \approx 0.97$), whereas subcooled performance was more variable. These findings suggest that regime-specific reporting can improve the robustness of HTC models derived from flow boiling data.